

TECHNICAL INTEGRATION & API HANDOVER

DOVA CHAIN — TECHNICAL INTEGRATION, API & AI CONNECTION DOCUMENT

Website · Backend API · AI Agent · Authentication · Database ·
E-Commerce Flow · Payments · Technical TODOs

REST API

JWT AUTH

AI AGENT

DEVELOPER HANDOVER

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Website: <https://dova.dntech.id/>

API Base URL: <https://api.dova.dntech.id/api/v1>

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01

PROJECT OVERVIEW

Project	DOVA Chain
Website	https://dova.dntech.id/
API Base URL	https://api.dova.dntech.id/api/v1

DOVA Chain is a digital food marketplace / supply-chain platform where customers can discover products, add products to a cart, create orders, make payments, and track orders through an AI-assisted shopping experience. The AI agent acts as the customer-facing interface, translating natural-language requests into structured calls against the DOVA Chain REST API.

02

SYSTEM ARCHITECTURE



Layer	Responsibility
Customer	Initiates requests in natural language via the website / AI chat surface.
Website / AI	Presents the shopping interface and routes customer intent to the AI agent.
AI Agent	Interprets intent, selects the correct HTTP tool, and enforces the order safety rules.
Custom HTTP Tools	Approximately 10 defined tools that translate AI intent into REST calls.
JWT Authentication	Secures protected endpoints via Bearer token validation.
REST API	Authoritative backend service exposing product, cart, order, and payment operations.
Database	Persists users, products, suppliers, carts, orders, payments, and delivery data.

03

WEBSITE CONNECTION

Website URL	https://dova.dntech.id/
Website Purpose	Customer-facing storefront and AI-assisted shopping entry point for DOVA Chain.
Frontend Status	REQUIRES VERIFICATION
Frontend Technology	Not detectable from current configuration — technician should verify.
Communication Method	HTTPS / REST API communication with the backend.
HTTPS Connection	CONFIRMED — site is served over HTTPS.
API Base URL Used	https://api.dova.dntech.id/api/v1
Authentication Dependency	Protected operations (cart, orders, payments) depend on a valid JWT Bearer token.

NOTE ON FRONTEND STACK

The exact frontend framework/technology powering https://dova.dntech.id/ is not confirmed in this document. Do not assume a specific framework — the technician should inspect the deployed site directly to confirm.

The frontend and/or AI layer should communicate with the backend exclusively over HTTPS, using the REST API base URL above, attaching a JWT Bearer token on any request to a protected endpoint.

04 BACKEND / API

Base URL	https://api.dova.dntech.id/api/v1
Protocol	HTTPS
Architecture	REST API
Authentication	JWT Bearer Token for protected endpoints

API Endpoint Reference

Method	Endpoint	Purpose	Authentication	Required Data
GET	/products	Search / list products	Document actual requirement	search / query
POST	/cart/add	Add product to cart	JWT	productId, quantity, deliverySlot
POST	/orders	Create order	JWT	deliveryName, deliveryAddress, deliveryPhone, fulfillmentType
GET	/orders	Retrieve order history / status	JWT	As required
POST	/payments/initialize	Initialize payment	JWT	orderId, amount
GET	/payments/verify	Verify payment	JWT / confirm actual requirement	reference

REGISTRATION / AUTHENTICATION ENDPOINTS

Exact paths must be verified from the backend API documentation. Do not invent endpoint names.

05 AUTHENTICATION

DOVA Chain uses a JWT (JSON Web Token) authentication architecture. The following sequence documents the expected flow — exact endpoint paths and token parameters must still be confirmed with the backend developer (see Section 18).

- Customer registration
- OTP generation
- OTP verification
- Account creation / login
- JWT issuance
- JWT storage / session
- Sending the token to protected endpoints
- Token expiration
- Session recovery
- Logout / session invalidation, if supported

Expected HTTP Concept

Authorization: Bearer <JWT_TOKEN>

CREDENTIAL SAFETY — MANDATORY

Never place a real secret, API key, JWT token, password, database password, or private credential inside this document. Use placeholders only: <JWT_TOKEN> , <API_SECRET> , <DATABASE_CREDENTIALS> .

06 AI AGENT CONNECTION

The DOVA Chain AI currently communicates with the backend through approximately 10 custom HTTP tools. The AI uses the shopping-and-orders playbook to determine when each tool should be called.

AI
↓
TOOL SELECTION
↓
HTTP REQUEST
↓
DOVA CHAIN API
↓
JSON RESPONSE
↓
AI INTERPRETATION
↓
CUSTOMER RESPONSE

07 IMPORTANT AI TOOLS

Product Search Tool

Calls	GET /products?search={{query}}
Purpose	Search the DOVA Chain product catalogue.
Input	"query" — e.g. "rice", "palm oil", "milk", "vegetables", "food"
Output	Product JSON objects containing available information such as productId, name, supplier, price, stock, category.

Add to Cart Tool

Calls	POST /cart/add
Required Info	productId, quantity, deliverySlot, authenticated customer / session
Rule	Do not add a product to the cart until the customer's requested quantity is clear.

Create Order Tool

Calls	POST /orders
Required Info	deliveryName, deliveryAddress, deliveryPhone, fulfillmentType
Rule	Do not create an order until the cart and delivery details have been confirmed.

Payment Initialization Tool

Calls	POST /payments/initialize
Required Info	orderId, amount

CRITICAL SAFETY RULE

The AI must only initialize payment after the backend successfully returns a valid Order ID.

Payment Verification Tool

Calls	GET /payments/verify
Uses	"reference"
Purpose	Verify whether the payment was actually successful.

RULE

The AI must never claim that payment is successful merely because payment initialization was successful.

Order Tracking Tool

Calls	GET /orders
Purpose	Retrieve order status / history.

The AI must use the backend response as the source of truth.

08 PRODUCT CATALOGUE — SOURCE OF TRUTH

The DOVA Chain API is the authoritative source for: product availability, product price, product stock, supplier, category, and product details. The AI must NOT invent product information.

Search Endpoint

```
GET https://api.dova.dntech.id/api/v1/products?search={{query}}
```

The `{{query}}` variable represents the customer's search term, passed through to the product search endpoint.

If No Results Are Returned

- Do not claim that DOVA does not sell the category.
- Try broader search terminology where appropriate.
- Inform the customer that no matching products were found only if broader searches also fail.

09 COMPLETE CUSTOMER ORDER FLOW

STEP 1 – Discovery

Customer asks for a product. → AI calls Search Products API

STEP 2 – Product Selection

AI displays products using API information.

STEP 3 – Registration / Login

If authentication is required: AI → OTP → Registration/Login → JWT

STEP 4 – Quantity Confirmation

Customer selects quantity.

STEP 5 – Add to Cart

AI → POST /cart/add

STEP 6 – Cart Confirmation

AI confirms the selected products and quantities.

STEP 7 – Delivery Details

Collect: Name, Address, Phone, Fulfillment type, Delivery slot (where applicable)

STEP 8 – Create Order

AI → POST /orders – Backend returns Order ID.

STEP 9 – Initialize Payment

AI → POST /payments/initialize – Only after successful Order ID creation.

STEP 10 – Payment Verification

AI → GET /payments/verify?reference=...

STEP 11 – Order Tracking

AI → GET /orders

10 ORDER SAFETY RULE

THE AI MUST NEVER:

- Create an order without customer confirmation.
- Invent an Order ID.
- Initialize payment without a valid Order ID.
- Claim payment was successful without payment verification.
- Invent product price.
- Invent stock availability.
- Invent delivery status.
- Claim an order exists if the API did not return one.

All transactional information must come from the backend API.

11 DATABASE ARCHITECTURE

INFERRED — MUST BE VERIFIED BY BACKEND DEVELOPER

The relationships below are inferred from the API's functional surface and are not confirmed against an actual schema.

Domain	Inferred Relationship
Users & Sessions	users → user_sessions
Catalogue	categories → products → supplier_profiles
Commerce	carts → cart_items → orders → order_items
Financial	payment_logs
Operations	stock_adjustments
Customer Communication	contact_submissions

Database engine: PostgreSQL / MySQL / etc. — UNCONFIRMED . Do not claim the exact database engine unless confirmed.

12 WHAT IS ALREADY CONNECTED

Item	Status
DOVA Chain website	Connected / Requires Verification
Backend API connectivity	Connected / Requires Verification
HTTPS communication	Connected
REST API architecture	Connected
JWT Bearer authentication	Connected / Requires Verification
Product search	Connected / Requires Verification
Product catalogue retrieval	Connected / Requires Verification
Cart functionality	Connected / Requires Verification
Order creation	Connected / Requires Verification
Order retrieval / tracking	Connected / Requires Verification
Payment initialization	Connected / Requires Verification
Payment verification	Connected / Requires Verification
Customer registration / OTP flow	Connected / Requires Verification
AI custom HTTP tools	Connected
Shopping-and-orders playbook	Connected
API source-of-truth logic	Connected
Order safety rules	Connected

Priority 1 — Session Validator

Current issue: There is no dedicated tool to verify whether the customer's authentication/session is active before attempting protected cart operations.

Required solution: Create a session validation / authentication-status mechanism. The AI should be able to determine:

- Is the user authenticated?
- Is the JWT valid?
- Has the token expired?
- Does the session still exist?
- Should the user authenticate again?

Reference: Todo "1bpt7p"

Priority 2 — Delivery Zone Validation

Current issue: There is no confirmed API-driven mechanism to determine whether a customer's address is inside the supported delivery zone.

Required solution: Create a delivery-zone validation endpoint/tool.

POST /delivery/validate-zone (conceptual example only)

IMPORTANT

This endpoint is only an example and must NOT be implemented under this exact path unless confirmed by the backend developer. The actual endpoint must be documented by the API owner.

The tool should determine: supported location, delivery availability, estimated delivery timeframe, next-day eligibility, delivery restrictions, and possible delivery fee if applicable.

Reference: Todo "rq5lce"

Priority 3 — Dynamic Error Handling

Current issue: Some failure messages are hardcoded in the AI instructions instead of being dynamically interpreted from API responses.

Required solution: The AI should read actual HTTP/API errors. Expected categories:

Status	Meaning
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
409	Conflict
422	Validation Error
429	Rate Limited
500	Server Error

The technician should map backend errors into safe customer-facing messages. Do not expose internal technical details to customers.

14 API ERROR & RESPONSE HANDLING

Every HTTP tool should capture:

- HTTP status
- Response body
- Error code if provided
- Message
- Request ID / correlation ID if available
- Relevant data

The AI should distinguish between the following response classes:

Class	Class
SUCCESS	NOT FOUND
PARTIAL SUCCESS	PAYMENT ERROR
VALIDATION ERROR	SERVER ERROR
AUTHENTICATION ERROR	NETWORK / TIMEOUT ERROR
AUTHORIZATION ERROR	

15 ENVIRONMENT & SECURITY

NEVER EXPOSE

- JWT secrets
- API secrets
- Database credentials
- Payment secret keys
- OTP secrets
- Encryption keys
- Admin credentials

Recommended Environment Separation

Environment	Purpose
Development	Active feature development and local testing.
Staging	Pre-production verification and QA.
Production	Live customer-facing environment.

Production credentials must be stored securely using environment variables or a secret-management system.

16 PAYMENT INTEGRATION

```
Customer confirms order
↓
Order created
↓
Order ID received
↓
Payment initialized
↓
Payment reference received
↓
Customer completes payment
↓
Payment verification request
↓
Backend confirms payment status
↓
AI reports verified result
```

IMPORTANT

Payment initialization ≠ successful payment. Payment is only considered successful when the verification endpoint confirms it.

Authentication Test

- ☐ Register customer
- ☐ Send OTP
- ☐ Verify OTP
- ☐ Receive JWT
- ☐ Test authenticated request
- ☐ Test expired token
- ☐ Test invalid token

Product Test

- ☐ Search product
- ☐ Search category
- ☐ Search nonexistent product
- ☐ Confirm price
- ☐ Confirm stock
- ☐ Confirm supplier

Cart Test

- ☐ Add product
- ☐ Add quantity
- ☐ Invalid quantity
- ☐ Insufficient stock
- ☐ Missing JWT
- ☐ Expired JWT

Order Test

- ☐ Create valid order
- ☐ Missing address
- ☐ Invalid phone
- ☐ Invalid fulfillment type
- ☐ Confirm Order ID returned

Payment Test

- ☐ Initialize payment
- ☐ Verify payment
- ☐ Invalid reference
- ☐ Failed payment
- ☐ Pending payment
- ☐ Successful payment

Delivery Test

- ☐ Supported address
- ☐ Unsupported address
- ☐ Next-day eligible address
- ☐ Non-eligible address

Tracking Test

- ☐ Valid Order ID
- ☐ Existing order
- ☐ Nonexistent order
- ☐ Different order statuses

18 REQUIRED INFORMATION FROM BACKEND DEVELOPER

- ☐ Complete API documentation
- ☐ Exact registration endpoints
- ☐ Exact OTP endpoints
- ☐ Login / session endpoint
- ☐ Session validation endpoint
- ☐ JWT expiration period
- ☐ Refresh-token mechanism, if available
- ☐ Complete list of all 10 AI HTTP tools
- ☐ Request / response schemas
- ☐ HTTP status codes
- ☐ Error-response structure
- ☐ Delivery-zone endpoint
- ☐ Delivery-slot rules
- ☐ Fulfillment-type values
- ☐ Payment provider documentation
- ☐ Payment webhook configuration
- ☐ Payment verification rules
- ☐ Database engine confirmation
- ☐ Database schema confirmation
- ☐ Production / staging environments
- ☐ API rate limits
- ☐ API versioning policy
- ☐ Logging / monitoring method

19 ENVIRONMENT VARIABLES

```
DOVA_API_BASE_URL=https://api.dova.dntech.id/api/v1
DOVA_JWT_TOKEN=<SECURE_VALUE>
DOVA_API_KEY=<SECURE_VALUE_IF_REQUIRED>
PAYMENT_SECRET=<SECURE_VALUE>
DATABASE_URL=<SECURE_VALUE>
```

WARNING

Never place production secrets directly inside source code, AI prompts, public GitHub repositories, screenshots, or documentation shared publicly.

20 DOVA CHAIN AI — NON-NEGOTIABLE RULES

1. Use the DOVA Chain API as the source of truth.
2. Never invent products.
3. Never invent prices.
4. Never invent stock.
5. Never invent suppliers.
6. Never invent Order IDs.
7. Never claim an order was created unless the API confirms it.
8. Never claim payment succeeded unless payment verification confirms it.
9. Never add products to a customer's cart without the required information.
10. Never create an order without the required delivery information.
11. Never expose JWTs or secrets to customers.
12. If authentication fails, guide the customer through authentication.
13. If delivery eligibility cannot be verified, do not falsely promise delivery.
14. If an API request fails, interpret the actual error where possible.
15. Do not use hardcoded product information when live API data is available.

21 TECHNICAL CONNECTION MAP

DOVA CHAIN WEBSITE

↓ [HTTPS](#)

DOVA CHAIN AI

↓ [Custom HTTP Tools](#)

AUTHENTICATION

↓ [JWT Bearer Token](#)

DOVA CHAIN API — <https://api.dova.dntech.id/api/v1>

↓

PRODUCT SERVICE — [Products](#) / [Categories](#) / [Suppliers](#)

↓

COMMERCE SERVICE — [Cart](#) / [Orders](#) / [Order Items](#)

↓

PAYMENT SERVICE — [Initialize](#) / [Verify](#) / [Payment Logs](#)

↓

DELIVERY — [Delivery Zone](#) / [Delivery Slot](#) / [Fulfillment](#)

↓

DATABASE — [Users](#) / [Products](#) / [Suppliers](#) / [Carts](#) / [Orders](#) / [Payments](#)

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CURRENT STATUS SUMMARY

Component	Current Status	Action
Website	Connected	Verify
REST API	Connected	Verify
HTTPS	Connected	Maintain
JWT	Connected	Verify implementation
Product Search	Connected	Test
Cart	Connected	Test authentication
Orders	Connected	Test
Payments	Connected	Test verification
Order Tracking	Connected	Test
Session Validator	Missing	Build
Delivery Zone Validation	Missing	Build
Dynamic Error Handling	Needs improvement	Implement
API Documentation	Partial	Complete
Database Schema	Inferred	Verify
Production Security	Needs verification	Audit

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FINAL DEVELOPER HANDOVER

DOVA Chain Technical Handover Objective

The goal is to ensure that the DOVA Chain website and AI agent operate as a reliable, secure, API-driven commerce system.

The Technician's Primary Objectives

- Verify every existing API connection.
- Verify authentication and session handling.
- Verify all AI HTTP tools.
- Verify product search and catalogue accuracy.
- Verify cart and order creation.
- Verify payment initialization and payment verification.
- Implement session validation.
- Implement delivery-zone validation.
- Replace hardcoded technical failure logic with dynamic API error handling.
- Complete API documentation.
- Perform end-to-end testing.
- Perform a security review before production launch.

Document Legend

Confirmed

 = verified in this document

Inferred

 = assumed from functional behavior, not verified

Requires Verification

 = technician must confirm

Required from Backend Dev

 = not yet available